Stay on the leading edge of stem cell research

The Gibco™ 5 Days of Stem Cells virtual event connects you to the latest stem cell techniques, research breakthroughs, and esteemed scientists from around the world–all from the comfort of anywhere

The 5-day virtual agenda is packed full of incredible insights in the form of:

- Leading scientific presentations from thought leaders around the world
- Behind-the-scenes virtual training demos
- Scientific poster sessions
- Hundreds of key stem cell tools and resources
- A global network of researchers including our stem cell experts and technical support

We'll share developments, discoveries, and cutting-edge content connected to a wide variety of stem cell applications, including disease modeling, cell and gene therapy, 3D modeling, and much more.

It's all happening October 12–16, 2020.

CURRENT PROTOCOLS
A Wiley Brand

Register for free at **thermofisher.com/5daysofstemcells**

ThermoFisher SCIENTIFIC

Received: 25 June 2019 | Revised: 23 October 2019 | Accepted: 22 November 2019
DOI: 10.1002/bit.27235

ARTICLE

BIOTECHNOLOGY and BIOENGINEERING WILEY

Process intensification by frontal chromatography: Performance comparison of resin and membrane adsorber for monovalent antibody aggregate removal

Sebastian Vogg | Felix Pfeifer | Nicole Ulmer | Massimo Morbidelli

Department of Chemistry and Applied Biosciences, Institute for Chemical and Bioengineering, ETH Zurich, Zurich, Switzerland

Correspondence
Massimo Morbidelli, Dipartimento di Chimica, Materialie Ingegneria Chimica, Politecnico di Milano, Via Luigi Mancinelli 7, 20131 Milano, Italy.
Email: massimo.morbidelli@polimi.it

Funding information
Kommission für Technologie und Innovation, Grant/Award Number: 19190.2 PFIW-IW

Abstract

Aggregates are amongst the most important product-related impurities to be removed during the downstream processing of antibodies due to their potential immunogenicity. Traditional operations use cation-exchange resins in bind-elute mode for their separation. However, frontal analysis is emerging as an alternative. In this study, a three-step process development for a membrane adsorber and a resin material is carried out, allowing the comparison between the stationary phases. Based on a screening study, optimal loading conditions are determined, which show that weak binding is favored on the membrane and strong binding on the resin. Transfer of these findings to breakthrough experiments shows that at 99% pool purity the yield is higher for the membrane, while the resin can be loaded twice as high, exceeding yields of 85%. For the investigated antibody and based on a given regeneration protocol, the productivity of the two phases is similar, ranging around 200 g/(L·h). Due to the higher loading, the resin requires about one-third less buffer than the membrane. Furthermore, the implementation of a wash step after loading allows to further increase yield by about 5%. In comparison to a generic bind-elute process, productivity and buffer consumption are improved by an order of magnitude.

KEYWORDS
aggregate, antibody, frontal chromatography, membrane adsorber, resin

1 | INTRODUCTION

The increasing diversity of therapeutic modalities in the pipelines of biopharmaceutical companies as well as growing competition in the market require the industry to intensify their processes to enable smaller, more flexible, and cost-effective unit operations (Jagschies, Lindskog, Lacki, & Galliher, 2018). In the case of downstream processing, chromatography remains the separation process of choice. The purification of monoclonal antibodies is typically carried out in three steps, to capture and subsequently polish the target molecule (Carta & Jungbauer, 2010). In polishing applications, cation-exchange chromatography (CEX) is usually applied for aggregate and fragment removal, which are the most abundant groups of impurities. Most fragments originate from protease activity during fermentation and subsequent harvest storage (Vlasak & Ionescu, 2011), and can therefore be largely reduced by decreasing residence time in the bioreactor (e.g., perfusion processes) and immediately capturing the resulting harvest (Karst, Steinebach, & Morbidelli, 2018). While the formation of fragments can therefore be greatly reduced by changing the fermentation process, aggregates are formed not only during upstream processing. For example, low-pH treatments encountered in the elution from protein A resins as well as in virus inactivation (VI) lead to aggregation (Arosio, Barolo, Müller-Späth, Wu, & Morbidelli, 2011; Arosio, Rima, & Morbidelli, 2013; Hari, Lau, Razinkov, Chen, & Latypov, 2010; Liu, Ma, Winter, & Bayer, 2010). As high molecular weight species (HMW) may pose a significant concern regarding the drug's efficacy and safety, they need to be reliably cleared down to low levels and controlled within tight specifications (Carta & Jungbauer, 2010). Traditionally, bind-elute

CEX processes are employed, where both impurities and products bind strongly to the stationary phase during loading and are then differentially desorbed by gradually changing buffer conditions towards higher ionic strength or pH. Consequently, only a fraction of the binding capacity can be utilized, reducing process productivity. As an alternative, frontal chromatography has been proposed to resolve monomers form dimers (Ichihara, Ito, Kurisu, Galipeau, & Gillespie, 2018; Liu et al., 2011; Reck, Pabst, Hunter, & Carta, 2017; Stone, Cotoni, & Stoner, 2019; Wollacott et al., 2015). Hereby, the feed material is applied to the stationary phase beyond its dynamic binding capacity for the monomer, allowing product collection during loading. As aggregates bind more strongly to the stationary phase, they elute later than the target product, and furthermore displace it on the surface. The displacement effect allows for high yields, even on stationary phases with substantial binding capacity for the monomer.

The purity constraint is relevant for the collected product, rather than the instantaneous purity at the outlet. Therefore, the initial part of the breakthrough curve with purity higher than 99% allows to continue collecting products even after the instantaneous impurity level at the outlet exceeds 1%. Before the product pool becomes contaminated with unacceptable levels of aggregates, loading is stopped, the stationary phase washed to desorb and recover monomer and stripped to remove bound aggregates. With respect to bind-elute processes, two effects lead to better utilization of the stationary phase: First, rather than a small fraction, almost the entire capacity can be utilized, which—second—is mainly done by the less abundant impurities. This is expected to increase throughput and therefore productivity.

In contrast to bind-elute chromatography, where a high number of theoretical plates is necessary to resolve the solutes during gradient elution, frontal chromatography does not have this requirement. Hence, membrane adsorbers are a suitable alternative to resin materials. Furthermore, they allow high flow rates due to their short bed height. Because of their convective nature, diffusion limitation encountered in porous beads is avoided and high flow rates do not result in shallower breakthrough curves. However, membrane adsorbers exhibit a smaller binding capacity as their specific surface area is smaller in comparison to resin materials. Consequently, membranes have been studied for protein purification using frontal analysis (Brown, Bill, Tully, Radhamohan, & Dowd, 2010; Liu et al., 2011; Weinbrenner & Etzel, 1994; Zhou & Tressel, 2006), although not yet under optimized conditions for antibody aggregate removal. More work was dedicated to resin materials addressing separation of various protein mixtures (Fargues, Bailly, & Grevillot, 1998; Garke, Hartmann, Papamichael, Deckwer, & Anspach, 1999; Tao, Chen, Carta, Ferreira, & Robbins, 2012) and also aggregate removal (Hunter & Carta, 2001; Liu et al., 2011; Wollacott et al., 2015), partially including model development (Reck et al., 2017). Furthermore, while resin materials traditionally have been optimized for overall binding capacity with bind-elute processes in mind, frontal chromatography for the removal of aggregates requires the selective retention of HMW species, while binding capacity for the monomer should be low. Recently (Hargreaves, 2018), a new resin material designed for this purpose was introduced showing promising results in the frontal analysis (Ichihara et al., 2018; Stone et al., 2019).

The main objective of this study is the experimental comparison of a membrane adsorber and a column operated in frontal mode. The comparison is made with respect to various process performance parameters and further discussed in light of the subsequent operations. Process development is carried out in a three-step approach, allowing a fair comparison between the stationary phases at their respective optimum. Optimal feed conditions in terms of pH and conductivity are determined for each stationary phase. These are subsequently transferred to dynamic experiments to define load amount and flow rate suitable for aggregate removal. Based on those, the effect of the wash conditions on the process outcome is evaluated.

2 | MATERIALS AND METHODS

2.1 | Feed generation

Cell-free supernatant containing a monovalent antibody (100 kDa, isoelectric point [pI] 7.3) in concentrations varying between 0.19 g/L and 0.54 g/L was obtained from perfusion cultivations of Chinese hamster ovary (CHO) cells as described in previous publications (Wolf et al., 2019). The antibody was captured on MabSelect SuRe (GE Healthcare, Uppsala, Sweden) using the CaptureSMB process and two prepacked columns (1.13 cm × 5 cm; Repligen GmbH, Ravensburg, Germany) on a Contichrom Cube (ChromaCon AG, Zurich, Switzerland) as described in Vogg, Wolf, and Morbidelli (2018) using 50 mM citrate, pH 3.5 for elution. The resulting eluate (3 mS/cm) was held at low pH and subsequently neutralized by adding 1 M Tris, pH 8.0 (10% by volume). Variation of the low-pH hold duration (0–120 min) resulted in different aggregate levels. Dimer and higher-order aggregate content consistently showed a ratio of approximately 2 to 1. Eluates were pooled to obtain desired HMW content and finally filtered through a 0.2-µm cut-off filter („rapid“-Filtermax; TPP, Trasadingen, Switzerland).

2.2 | Stationary phases

Sartobind S (Sartorius AG, Göttingen, Germany) was obtained as 8-strips (19 µl membrane volume) compatible with 96-well plates for the screening study. For the dynamic experiments, a 3-ml capsule with 8-mm bed height was used. The capsule was regenerated for all experiments showing consistent performance throughout. Eshmuno CP-FT (Merck KGaA, Darmstadt, Germany) was obtained in bulk and prepacked in 0.2-ml column (0.5 × 1 cm; Superformance) by Merck. The column was regenerated as the membrane.

2.3 | Buffer system

Process development and performance characterization of the two stationary phases were done in the context of a lean process, which would be integrated to previous and subsequent unit operations. Therefore, buffer exchange between these unit operations is sought to be minimized. When placing the CEX polishing step directly downstream of capture and VI, a mixture of protein A elution buffer and neutralizing agent is subjected to the CEX step. As the capture step elution buffer for this antibody consists of citric acid, and phosphate is well suited for neutralization, the buffer that constituted of the mixture of the two was selected.

Buffers at different pH and conductivity were obtained by mixing 50 mM citric acid (Fisher Scientific, Loughborough, UK), pH 3.5 with 250 mM sodium phosphate (Sigma–Aldrich, Steinheim, Germany), pH 8.0 and MilliQ water. The set points of each buffer are summarized in Table S1.

2.4 | Screening study

Buffer exchange was carried out as follows: Amicon Ultra-15 ultrafiltration spin tubes (Merck KGaA) with a molecular weight cut-off (MWCO) of 10 kDa were loaded with 10 ml of protein A eluate and centrifuged for 10 min at 5,000*g* (Centrifuge 5430; Eppendorf, Hamburg, Germany). Afterward, 10 ml of buffer was added and the tube was centrifuged at the same conditions as before. This step was repeated three times. Upon collection of the retentate, the protein pool was diluted to 5 g/L and analyzed confirming unchanged impurity profiles and targeted feed concentration. Final pH and conductivity are summarized in Table S1. Permeates were pooled and analyzed showing complete retention of the target protein in the retentate.

2.4.1 | Membrane adsorber

8-strips of Sartobind S were connected to 96-well plates (500 µl volume). After each loading, the setup was centrifuged at 1,000*g* for 5 min (Centrifuge 5810 R; Eppendorf). The setup was then rotated and centrifuged again, as the fluid level was found to sink nonhomogeneously across the different membranes. First, 500 µl of 30% ethanol in MilliQ water was used to wet the membrane. Second, it was equilibrated twice with 500 µl of buffer at the respective feed pH and conductivity. Third, the feed was applied such that the loadings of 50 g/L, 100 g/L, and 200 g/L were obtained. As the last exceeded the maximum volume of 500 µl, two loadings were carried out and the flow-through pooled. Each buffer condition and loading was evaluated in duplicate.

2.4.2 | Resin material

The bulk resin was centrifuged at 4,700*g* for 10 min (Heraeus Megafuge 16R; Thermo Fisher Scientific, Waltham, MA). The supernatant was removed and equilibration buffer at pH 4.5 and conductivity 4 mS/cm was added. The suspension was shaken on a platform shaker for 20 min before centrifuging it again. This step was repeated three times. Afterward, the resin pellet was suspended to yield a 50% slurry. While shaking, 20 µl of the slurry was transferred to 1.5 ml tubes. 1 mL of feedstock was added, and the dispersion was shaken for 20 min. Afterward, the tubes were centrifuged at 13,000*g* for 10 min (Centrifuge 5415 R; Eppendorf). The supernatant was carefully removed from each tube, filtered, and analyzed. Feedstock addition was repeated twice resulting in loadings of 500, 1,000, and 1,500 g/L. Each experiment was performed in duplicate.

2.5 | Breakthrough curves and washing protocol

Buffer exchange on a Sartoflow Slice 200 Benchtop system (Sartorius AG) equipped with a Sartocon Slice 200 Hydrosart membrane (10 kDa MWCO; Sartorius AG) was carried out as follows: After washing the membrane with the desired buffer, the eluate pool was concentrated two-fold and diafiltered against the desired buffer using at least five volume equivalents with respect to the protein pool. The retentate was diluted to yield a protein pool at around 5 g/L. The resulting feedstocks are summarized in Table 1. Permeates were analyzed proving complete retention of the target protein.

Frontal loading experiments were performed on a Contichrom CUBE (ChromaCon AG). For the breakthrough curves, the protocol for both stationary phases consisted of the same steps: After initial

TABLE 1 Properties of feedstocks used in dynamic experiments

	Set point		Feed properties			
Purification unit	pH	Conductivity, mS/cm	pH	Conductivity, mS/cm	Concentration, g/L	HMW (%)
Breakthrough curves						
Sartobind S	5.5	8	5.6	7.7	4.9	5.7
Sartobind S	5.5	10	5.6	10.1	4.9	5.5
Eshmuno CP-FT	5.0	4	5.1	3.9	4.9	5.7
Eshmuno CP-FT	5.5	4	5.5	3.8	4.7	5.2
Batches						
Sartobind S	5.5	8	5.5	7.8	5.0	12.0
Eshmuno CP-FT	5.0	4	5.0	4.0	4.5	4.9

Abbreviation: HMW, high molecular weight.

equilibration, loading was carried out followed by a wash step using equilibration buffer. Stripping was done using 1 M NaCl in equilibration buffer while 1 M NaOH was applied for cleaning-in-place (CIP) before re-equilibrating the stationary phase. Flow-through from the stationary phase was fractionated during loading, wash and strip, and afterward analyzed for the monomer and aggregate content. Flow rates and volumes applied for each step and stationary phase are summarized in Table S2. Flow-through during loading, wash, and strip was fractionated, filtered (0.2 µm), and analyzed.

For assessment of the impact of different wash conditions, the following approach was applied, which will be referred to as batch experiments. A safety margin of 15% with respect to the loading corresponding to 99% pool purity was used to define the feed amount. Loading and wash were carried out at the highest flow rate evaluated in the breakthrough experiments. Wash conditions were varied in terms of pH and conductivity. While the applied wash step consisted of larger volumes, only the first five column volumes/membrane volumes (CV/MV) of the wash were evaluated towards their impact on process outcome. Flow rates and volumes applied for each process step and stationary phase are summarized in Table S2.

2.6 | Performance evaluation

The screening study was evaluated in terms of purity and yield. Breakthrough experiments were evaluated furthermore with respect to productivity and buffer consumption. Purity was obtained directly from the analytical size-exclusion chromatogram as the ratio of monomer area and total area at 280 nm. Yield as a function of loading L was determined based on the cumulative mass of monomer collected during loading and wash from the stationary phase $m_{\text{coll},M}$ and the mass-loaded up to this point $m_{l,M}$:

$$Y(L) = \frac{m_{\text{coll},M}(L)}{m_{l,M}(L)}. \tag{1}$$

Productivity of a single cycle as a function of loading L was calculated as the ratio of cumulative mass of monomer collected and the product of volume of stationary phase V_S and cycle time t_C:

$$\text{Prod}(L) = \frac{m_{\text{coll},M}(L)}{V_S t_C(L)}. \tag{2}$$

Cycle time consists of the time required for loading t_L and the duration needed for replacement or regeneration of the stationary phase t_{RR}. In the following, the latter is also referred to as turnover time and includes washing, stripping, CIP, and re-equilibration steps. Yield and productivity of the batch experiments were based on the mass of monomer collected during loading and washing. Buffer consumption as a function of loading L was obtained as the ratio of required buffer volume and the cumulative mass of collected monomer:

$$BC(L) = \frac{V_{\text{Buffer}}}{m_{\text{coll},M}(L)}. \tag{3}$$

The mass balance evaluated for dynamic experiments deviated less than 3% for the monomer and less than 7% for aggregates, which can be attributed to the low content in many fractions and therefore noisy measurements.

2.7 | Analytics

Total protein concentration and aggregate content were determined by analytical size-exclusion chromatography using a TSKgel G3000SWXL column (0.78 cm × 30 cm; TOSOH Bioscience, Japan). Isocratic elution was carried out at 1 ml/min using 50 mM sodium phosphate, 300 mM NaCl, pH 7.0. The UV detector of an Agilent 1100 HPLC (Agilent, Santa Clara, CA) was calibrated with samples of known concentration at 280 nm.

3 | RESULTS AND DISCUSSION

3.1 | Screening study

The goal of the screening study as the first step in this process development was to establish optimal loading conditions in terms of pH and conductivity. Experiments under static conditions were carried out as neither pH nor ionic strength have an impact on mass transfer in the investigated range of operating conditions. Therefore, thermodynamic conditions optimally suited for the separation can be obtained from this study. The considered experimental conditions are summarized in Table S1.

3.1.1 | Membrane adsorber

Results of the screening study at 6% and 15% aggregates on Sartobind S are shown in Figure 1 for 200 g/L loading. Purity first increases with increasing conductivity and pH reaching a maximum in a region between pH 5.5, 10 mS/cm and pH 6.0, 6 mS/cm. In the case of the 6% HMW feedstock, purities above 99% can be achieved, while only 95% purity is possible at this loading in the case of 15% HMW in the feed. When increasing pH and conductivity further, purity decreases. The observed trends are consistent for the two different impurity contents. On the other hand, yield generally increases with pH and conductivity, reaching values above 85%. At very strong binding conditions (pH 4.5, 4 mS/cm), it still increases but to a much lower extent. Other evaluated loadings show consistent trends (Figures S1 and S2).

The outcomes can be rationalized by the following considerations: At low pH and conductivity, both monomer and aggregates show strong binding and high capacities resulting in low yield and relatively low resolution and therefore purity. When increasing pH and conductivity, the binding becomes significantly weaker for the monomer while dimers and larger aggregates are still strongly bound. Hence, purity and yield increase. At even higher pH and conductivity,

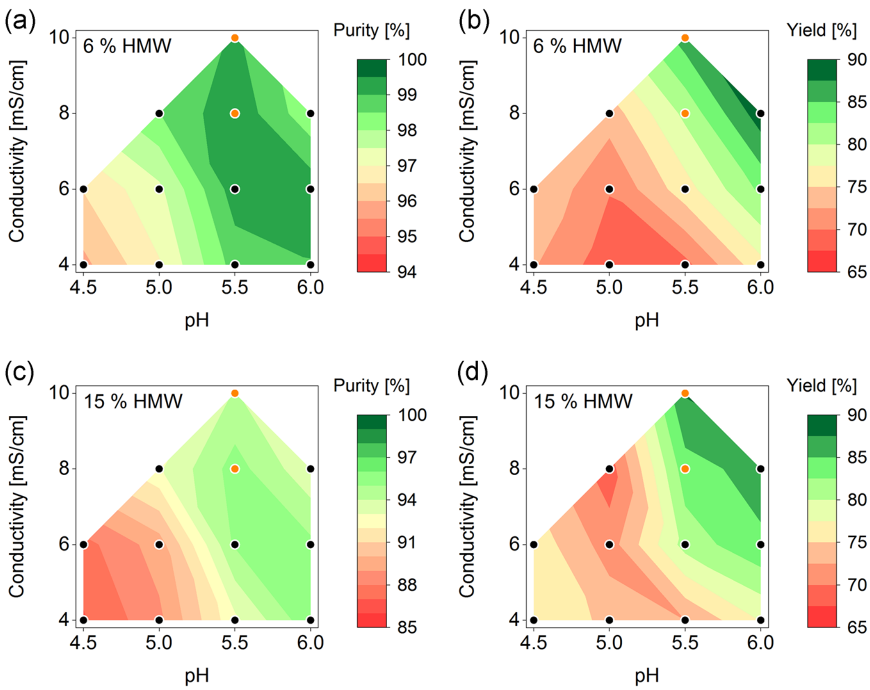

FIGURE 1 Purity (a,c) and yield (b,d) at 6% (*Top row*) and 15% (*Bottom row*) high molecular weight (HMW) content, as obtained from the screening study at 200 g/L loading on Sartobind S. Black circles indicate evaluated combinations of pH and conductivity. Orange circles designate the two best loading conditions, which were transferred to the breakthrough experiments [Color figure can be viewed at wileyonlinelibrary.com]

binding of aggregates also becomes weak, resulting in worse purity, while yield increases further due to less monomer retention. This behavior is in line with a previous study showing successful monomer purification only in a certain window of ionic and hence binding strength (Reck et al., 2017). Furthermore, the behavior of the yield at pH 4.5 should be mentioned, which shows higher values than pH 5.0 and can be attributed to the following considerations. As the binding is strong under these conditions, the equilibrium binding capacity is practically the saturation binding capacity (SBC) at the given conditions. As the SBC is a function of pH itself (Wrzosek & Polakovič, 2011), we conclude that SBC at pH 4.5 is lower than at pH 5.0 resulting in the observation of higher yields.

As a basis to define the best loading conditions, which are further investigated in dynamic breakthrough experiments, the process performance for each pH–conductivity combination is shown on the purity-yield plane as a function of loading in Figure 2a. No single set of conditions seemed to clearly outperform the others. Therefore, the two best pH–conductivity combinations were chosen for further investigation, that is, at pH 5.5 with 8 mS/cm and 10 mS/cm, respectively. The combination pH 5.5, 6 mS/cm was omitted due to the significantly lower yield compared to the other two combinations.

3.1.2 | Resin material

Results of the screening study at 6% and 15% aggregates on Eshmuno CP-FT are shown in Figure 3 for 500 g/L loading. Yield shows the same trend as for Sartobind S. It increases with decreasing binding strength. On the other hand, in terms of purity, the resin and the membrane show opposite behaviors. The purity is highest at low conductivity, which can be rationalized by different ligand design: While both stationary phases have a sulfonyl-type functionalization, charges are diluted and spaced out on the resin. Consequently, a lower number of interaction groups are accessible to the smaller monomer, compared to the aggregates. Therefore, retention is comparably low even at low conductivity. As aggregates remain to be retained strongly, the highest purity results for strong binding conditions. The observed tendencies are also found for higher loadings (Figures S3 and S4).

At 500 g/L loading, the resin achieves a maximum purity of 98.5% for the 6% HMW feedstock and of 92.5% for the 15% HMW feed. Optimal conditions are determined as for the membrane. They are found for 4 mS/cm at pH 5.0 and 5.5, respectively (Figure 2b). While this is different from the membrane's optimum, it can be

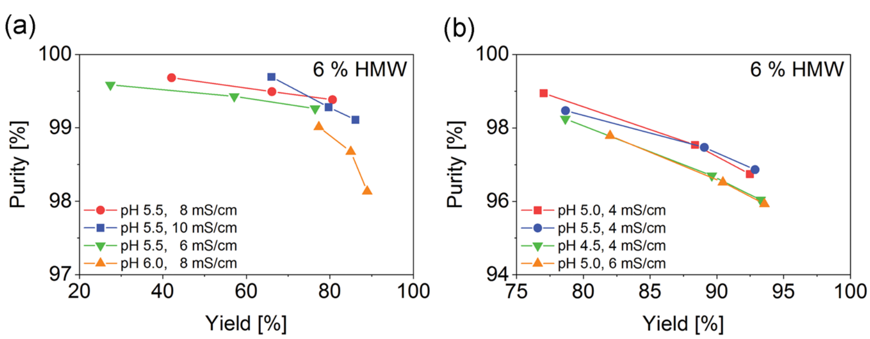

FIGURE 2 Purity as a function of yield as obtained from the screening study on Sartobind S (a) and Eshmuno CP-FT (b) for selected loading conditions at 6% high molecular weight (HMW) content. The lines connect results from different loadings for each pair of pH and conductivity [Color figure can be viewed at wileyonlinelibrary.com]

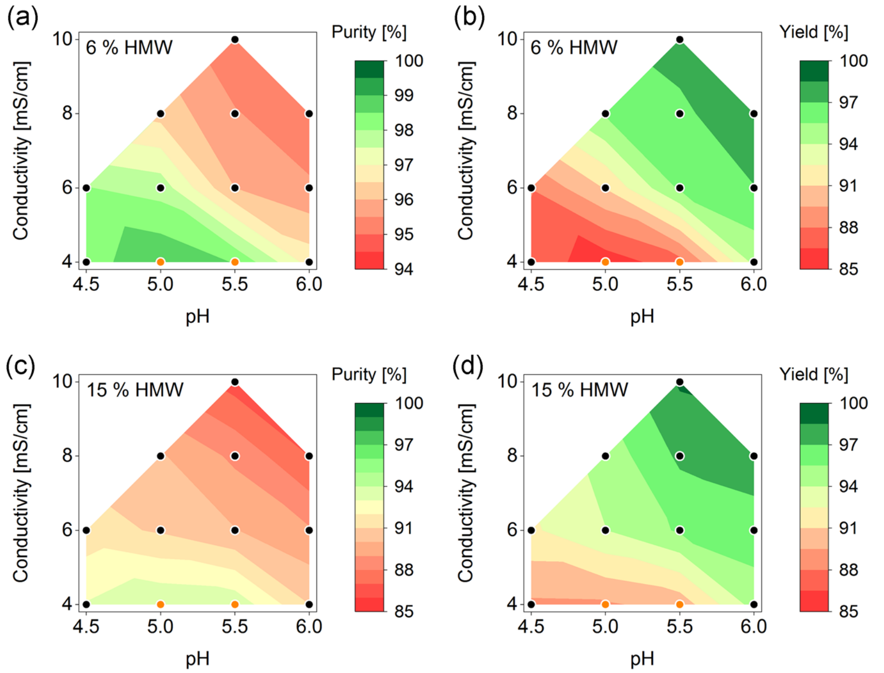

FIGURE 3 Purity (a,c) and yield (b,d) at 6% (*Top row*) and 15% (*Bottom row*) high molecular weight (HMW) content as obtained from the screening study at 500 g/L loading on Eshmuno CP-FT. Black circles indicate evaluated combinations of pH and conductivity. Orange circles designate the two best loading conditions, which were transferred to the breakthrough experiments [Color figure can be viewed at wileyonlinelibrary.com]

attributed to the different ligand chemistry of the two stationary phases as described above. Furthermore, the found optimum is about 2 pH units lower than the antibody's pI, which is in line with the findings of Ichihara et al. (2018) and Ichihara, Ito, and Gillespie (2019), and confirms this pH region to be a good starting point for screening studies. Conductivities lower than 4 mS/cm might yield even better results (Stone et al., 2019); however, these conditions were not investigated due to considerations regarding the entire downstream processing cascade. To achieve such low conductivities in fact, protein A eluate should have to be neutralized with low molarity, high pH buffers. For the given system, neutralization to pH 5.0, 4 mS/cm requires more than two-fold dilution. Reducing the conductivity further would therefore result in even lower protein concentrations and higher volumes to be handled. Both effects should be avoided.

Beyond the trends for purity and yield, a deeper comparison between the stationary phases is not possible at this point. This is due to the static nature of the experiments as well as the fact that a few distinct loading values were considered.

3.2 | Breakthrough curves

As the second step of process development, breakthrough experiments were performed at the two best combinations of loading pH and conductivity as determined in the screening study. The dynamic nature of these experiments allows a comparison between membrane and resin, since it involves also the mass transport characteristics of the two systems. As purity is the primary objective of the polishing steps, this was taken as a basis for the comparison. For aggregates, typically a level below 1% is targeted (Carta & Jungbauer, 2010), which was taken as a constraint in this study.

3.2.1 | Membrane adsorber

Breakthrough curves of monomer and aggregates on Sartobind S were obtained with feedstocks at pH 5.5 with 8 mS/cm (Figure 4a) and 10 mS/cm (Figure 4c), respectively. Residence times of 12 s, 30 s, and 1 min were evaluated. As expected from the convective nature of the membrane, the flow rate had an influence neither on monomer nor on aggregate breakthrough as a function of the loaded volume. Hence, convection remains the rate-limiting step not only with respect to film transport from bulk liquid to membrane surface and to binding kinetics of the monomer but also with regard to the kinetics of monomer displacement by aggregates. While the first two results are typical for membrane adsorbers (Knudsen, Fahrner, Xu, Norling, & Blank, 2001), the displacement kinetics can potentially be slower than single-component adsorption as shown for a resin material (Reck et al., 2017), and therefore possibly become rate-limiting. As shown in Figure 4, this is not the case for the system under investigation.

The breakthrough of the monomer occurs already at low loadings: 90% of the feed monomer concentration is reached after 35 g/L (8 mS/cm) and 20 g/L (10 mS/cm). The displacement effect is macroscopically not very pronounced as breakthrough levels exceed the feed concentration only slightly (8 mS/cm), if at all (10 mS/cm). However, the true flow-through behavior of the monomer can be precluded even for the 10 mS/cm case as wash and strip fractions contained monomer, corresponding to 14 g/L of bound monomer after 260 g/L loading. An aggregate breakthrough occurs much slower reaching values of 28% (8 mS/cm) and 70% (10 mS/cm) after 250 g/L loading. Pool purity as well as process yield were calculated from the breakthrough curves as shown in Figures 4b and 4d. Of course, purity decreases with loading while yield increases. At 10 mS/cm, a 99% pool purity is obtained at 85 g/L loading with a yield value of 90%. At 8 mS/cm, the membrane can be loaded beyond

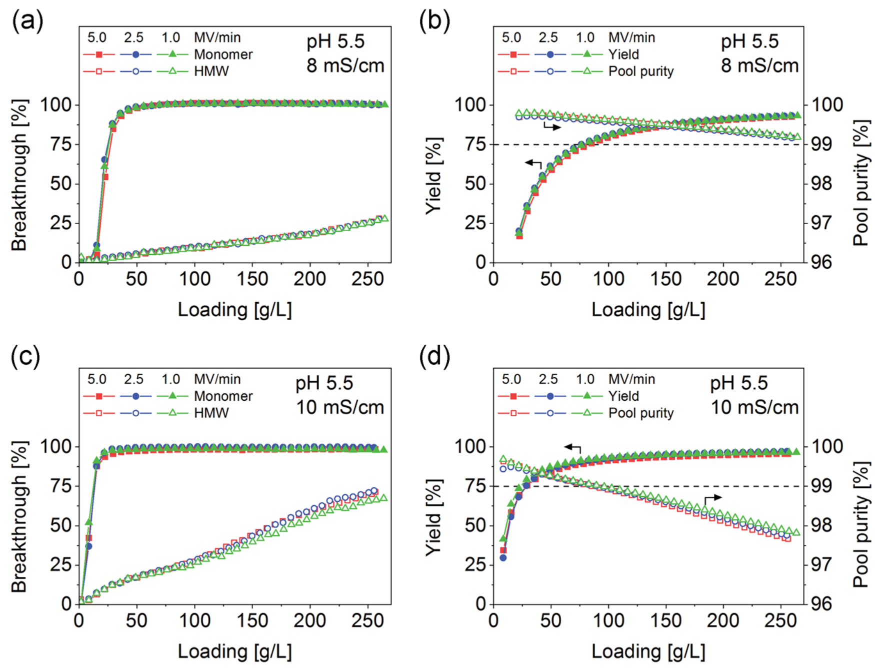

FIGURE 4 Breakthrough curves at different flow rates on Sartobind S (a,c) and corresponding pool purity and process yield (b,d) for feed conditions pH 5.5, 8 mS/cm (*Top row*) and pH 5.5, 10 mS/cm (*Bottom row*). Feedstock characteristics are summarized in Table 1 [Color figure can be viewed at wileyonlinelibrary.com]

260 g/L before violating the purity constraint resulting in a yield higher than 93%. Therefore, it is concluded that both feed conditions facilitate the removal of aggregates at elevated yields. However, due to the possibility to reach significantly higher loadings and slightly higher yield, feedstock at pH 5.5 and 8 mS/cm was further investigated in batch experiments as discussed below.

3.2.2 | Resin material

Breakthrough curves of monomer and aggregates on Eshmuno CP-FT were obtained at pH 5.0 (Figure 5a) and pH 5.5 (Figure 5c) with 4 mS/cm. Residence times of 30 s, 1 min, and 2 min were considered. Surprisingly, the flow rate did not have any impact on the separation performance. Due to the short bed length, it is hypothesized that this effect is due to axial dispersion outweighing the effect of interphase mass transport resistances, which typically prevail in the case of longer columns with less relative dead volume. Therefore, it can be expected that the typical flow-rate dependence can be observed on columns with larger bed height due to the microporosity of the resin resulting in intraparticle diffusion or even adsorption becoming the rate-limiting step. In this case, a detailed modeling approach would be required to elucidate the rate-limiting step, which is beyond the scope of this work. Nevertheless, the above hypothesis is supported by the very large value measured (data

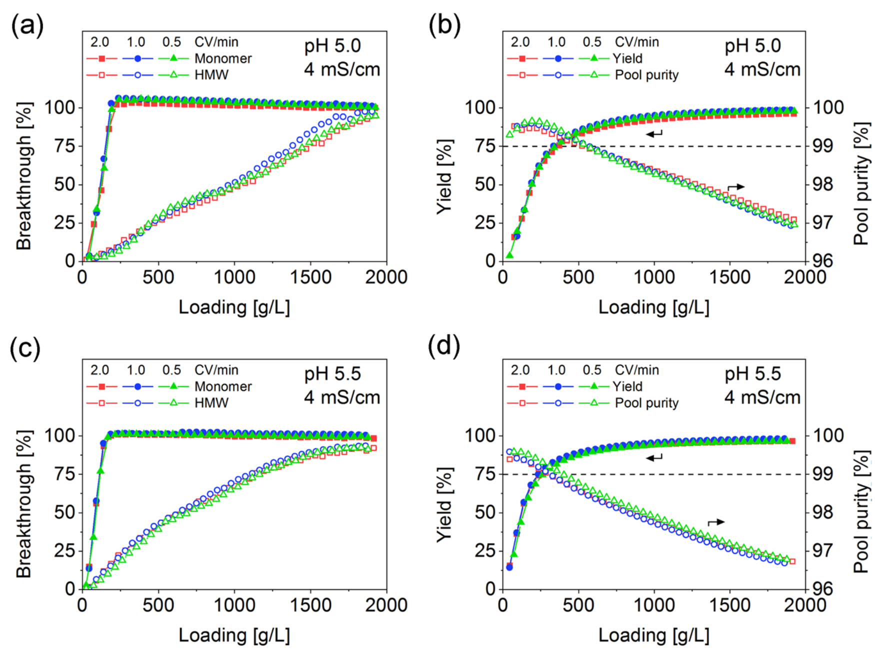

FIGURE 5 Breakthrough curves at different flow rates on Eshmuno CP-FT (a,c) and corresponding pool purity and process yield (b,d) for feed conditions pH 5.0, 4 mS/cm (*Top row*) and pH 5.5, 4 mS/cm (*Bottom row*). Feedstock characteristics are summarized in Table 1 [Color figure can be viewed at wileyonlinelibrary.com]

not shown) for the height equivalent to a theoretical plate, which in conjunction with the short bed height indicates a very strong impact of axial dispersion.

Nevertheless, separation remains successful and the data are meaningful for determining the most convenient conditions for the following process performance comparison study. The full monomer breakthrough is reached after ~200 g/L loading for both feed conditions. It is followed by a slowly decreasing overshoot indicating the presence of the displacement effect. Almost full breakthrough of aggregates is reached after 2,000 g/L loading. In the case of pH 5.5, the binding of aggregates is less strong due to closer proximity to the antibody's pI resulting in higher breakthrough earlier on. This also becomes apparent in terms of pool purity (Figures 5b and 5d). While the purity constraint is violated between 300 g/L and 350 g/L loading of feedstock at pH 5.5, feeding at pH 5.0 can be continued up to 530 g/L before reaching 99% pool purity. As discussed above, the breakthrough of the monomer is not strongly influenced by the feed pH, showing the expected influence on yield. Loading at pH 5.0 results in a yield of 86%, which is 5 percentage points greater than for the feedstock at pH 5.5. Therefore, loading conditions of pH 5.0 and 4 mS/cm were further investigated in batch experiments as discussed below.

3.3 | Performance comparison

The presented approach based on a screening study to obtain optimal loading pH and conductivity, and on breakthrough curve measurements to define loading amount and flow rate, allows for the quantitative comparison between the two purification units in terms of their process performance. While axial dispersion was dominating in the column, its effect becomes comparable to the one of mass transfer resistance at short residence times, which are investigated in this study. The productivity of each process, evaluated at the loading corresponding to 99% pool purity and at the highest considered loading flow rate in Figures 4 and 5, respectively, is shown as a function of the turnover time in Figure 6a. Turnover time is the time required for regeneration or replacement of the stationary phase. It is added to the loading duration to form a complete cycle. Due to its single-use nature, the membrane could be replaced after each loading. Since this also corresponds to unproductive downtime, it has been included in the turnover time (replacement time).

As discussed above, loading at pH 5.5, 8 mS/cm and pH 5.0, 4 mS/cm are the best conditions considered for the membrane and column process, respectively. For the membrane, a short turnover time results in very high productivity exceeding 1,000 g/(L·h). This is possible through the very high loading flow rate. With increasing turnover time, however, productivity decreases quickly as the total process duration is more and more determined by regeneration. This decrease is less pronounced in the case of the column, as a result of the lower flow rate and higher loading capacity, which also limit the maximum achievable productivity to about 470 g/(L·h). While such short turnover times are practically not attainable, at typical turnover times between 50 min and 80 min, both purification units show similar productivities at around 200 g/(L·h).

For further comparison, a regeneration protocol consisting of wash, strip (each 5 MV/CV), CIP (60 min), and re-equilibration (5 MV/CV) is considered. The wash step is carried out at the loading flow rate, while strip and re-equilibration are run at 5 MV/min and 1 CV/min for the membrane and column, respectively. Residence time during the CIP step was set to 5 min. This results in turnover times of 63 min (membrane) and 72.5 min (column), resulting in the membrane being slightly more productive than the column (Figure 6a). Buffer consumption was also evaluated resulting in 0.10 L/g for the membrane and a lower value of 0.06 L/g for the column due to the higher loading and therefore throughput per regeneration cycle.

More important than productivity and buffer consumption is probably yield, which is shown in Figure 6b as a function of productivity. Each curve corresponds to the process performance obtained by considering different loading times selected along the same breakthrough curve and therefore has changing pool purity. Safety margins are taken into consideration, which corresponds to decreasing loading, for example, a 10% safety margin corresponds to 90% of the loading at 99% pool purity. They are indicated by the

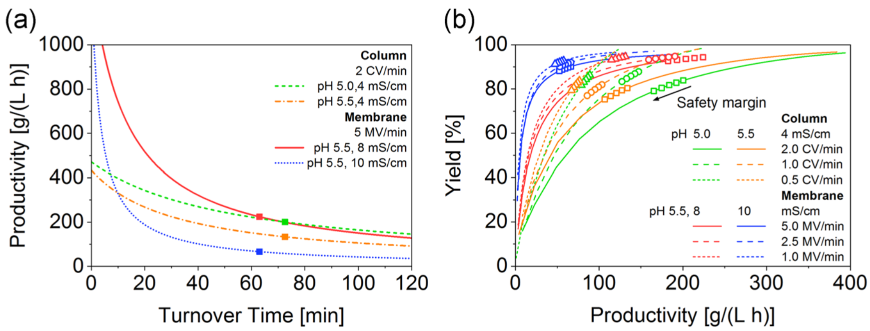

FIGURE 6 (a) Productivity as a function of turnover time for the two loading conditions considered for each purification unit and at the largest loading flow rate considered in Figures 4 and 5. Squares indicate the turnover time corresponding to the considered regeneration protocol. (b) Yield as a function of productivity. Each trajectory is constructed from a single breakthrough curve and has changing pool purity. Rightmost symbols on each line correspond to 99% pool purity, while the symbols to the left indicate the process performance upon application of a safety margin, incremented in steps of 5% [Color figure can be viewed at wileyonlinelibrary.com]

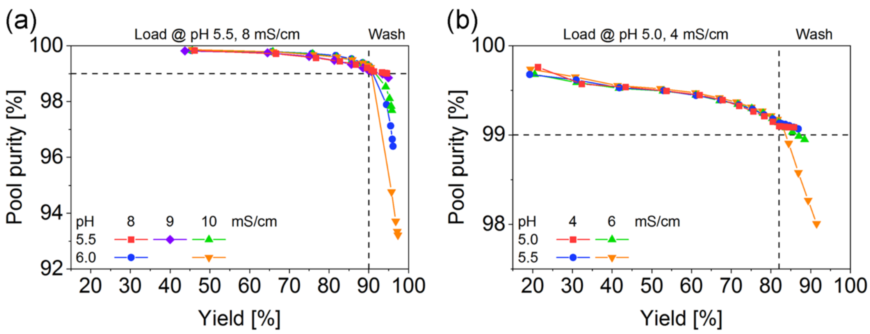

FIGURE 7 Influence of wash condition (pH and conductivity) on pool purity for the membrane (a) and the column process (b). Feed conditions are summarized in Table 1 [Color figure can be viewed at wileyonlinelibrary.com]

symbols on each curve: The rightmost on each line indicates 99% pool purity (0% safety margin), while the symbols to the left, correspond to increasing safety margins in increments of 5%. While both units achieve similarly high productivity, the yield is significantly higher for the membrane and furthermore it is less sensitive to the safety margin, which indicates higher process robustness. This is attributed to the weak binding conditions, at which the membrane is operated in conjunction with a displacement effect leaving only little monomer bound to the stationary phase at high loadings. It should also be pointed out that the absolute decrease of loading introduced by the safety margin is different for the two stationary phases. If compared on this absolute basis, both show approximately the same loss in yield upon decreasing the loading. Furthermore, it should be noted that this analysis does not include the possibility to design the wash step to recover the product left in the void volume or adsorbed on the surface, which is addressed in the next section. Nevertheless, as this step is incorporated in the regeneration protocol, the obtained values provide lower boundaries for achievable yield and productivity.

It is now interesting to compare frontal operation to a generic bind-elute process at two loadings of 40 g/L and 80 g/L. For regeneration, the same protocol as above was assumed for both processes. Loading at 1 CV/min with a feedstock at 5 g/L total protein concentration and 6% HMW content results in 8 CV and 16 CV feed, respectively. Gradient elution is set to 10 CV, also run at 1 CV/min. Assuming 100% yield, productivity attains values of 24 g/(L·h) and 45 g/(L·h), respectively. Therefore, frontal chromatography allows 5 to 10 times higher productivities than traditional bind-elute processing. Similarly, buffer consumption of 1.2 L/g and 0.7 L/g in the case of bind-elute operation can be reduced by an order of magnitude through frontal analysis. Although these values are strictly valid only for the system under consideration, it can indeed be concluded that frontal chromatography exhibits a great potential to intensify polishing operations for aggregate removal. Flow-through chromatography, that is, adsorption of impurities only, remains elusive for the separation of monomers and aggregates due to their similar properties and is therefore not considered in this comparison.

3.4 | Design of the wash step

In frontal chromatography operation, the wash step can be particularly important since it allows to recover the monomer left in the voids of the purification unit and to desorb at least a portion of the monomer bound to the solid, so as to increase yield, without however desorbing also the HMW, which would decrease purity. Accordingly, as the third step of the purification process development, we analyze the proper design of the wash step. Based on the respective breakthrough curve, a 15% safety margin with regard to the loading corresponding to 99% pool purity was applied (breakthrough data not shown). We would like to emphasize that a direct performance comparison between the stationary phases is not possible here as feedstocks with different HMW content were used for the batch experiments (Table 1). As expected, the breakthrough of monomer and aggregates is very reproducible (Figure S5).

Results in terms of pool purity and yield are shown in Figure 7 for various wash conditions. In the case of the membrane, yield after loading already reaches 90%, while pool purity is at 99.1% which allows for additional HMW elution before purity specification violation, which is set at 99%. While wash conditions at pH or conductivity higher than in the feed result in pool contamination, washing at feed conditions (pH 5.5, 8 mS/cm) allows to increase the yield to 94% without violating the purity specification. This results in productivity of 106 g/(L·h) based on the considered regeneration protocol.

Regarding the column, pool purity after loading attains a value of 99.1% with a yield of 82%. The two wash strategies with 4 mS/cm at pH 5.0 and pH 5.5, respectively, allow to increase yield without violating the purity constraint by 4–5 percentage points corresponding to 175 g/(L·h) productivity. As indicated above, the stationary phases cannot be compared against each other in this case as different feedstocks were applied. It should be noted that for even lower loadings more stringent wash conditions are expected to work as well. In this case, pool purity would be higher allowing to accommodate more aggregates before violating the purity specification.

4 | CONCLUSIONS

We applied a three-step procedure for the development of the purification process of an antibody from its aggregates using frontal CEX chromatography, achieving yields above 85% and productivities between 100 g/(L·h) and 200 g/(L·h), depending on initial HMW content. First, a screening study is used to establish optimal loading conditions in terms of pH and conductivity, followed by dynamic experiments establishing the load amount and flow rate. With the introduction of a safety margin, which reduces the loading by a certain percentage, a wash protocol can be defined in the third step to further increase yield, while remaining within the purity specification limits. It is shown that using feed pH and conductivity for the wash step provides a robust option when working with small safety margins.

It is worth noting that the presented comparison between the membrane adsorber and the column is fair because optimal loading conditions were developed for each phase independently, and the same feed material was applied in terms of concentration and impurity content. Of course, the presented approach does not result in a global optimum, which is accessible with reasonable experimental effort only using a model-based approach (Baur et al., 2018).

Summarizing, we can conclude that neither the membrane nor the resin clearly outperforms the other for the investigated target molecule. Aspects in favor of the membrane adsorber are the slightly higher productivity at typical turnover times of around 1 hr and, more importantly, the higher yield by about 10%. Furthermore, less dilution of protein A eluate is required as higher conductivity is necessary for good separation performance. On the other hand, subsequent further polishing such as anion-exchange in flow-through mode might suffer from the increased ionic strength requiring further dilution after the CEX step. By contrast, due to the low conductivity required for the resin material, dilution of protein A eluate is higher before the CEX step, but possibly avoids further buffer adjustment for subsequent polishing. In addition, very high productivity, lower buffer consumption, and well-established implementation at larger scale speak in favor of the resin. In any case, frontal chromatography in CEX mode is shown to be a lean and straight-through process for aggregate removal, enabling productivities 5–10 times higher than in traditional bind-elute operation. Based on the presented results, the decision on the purification unit is expected to be case-specific, including the molecule considered as well as the specific material in the membrane or in the column.

The effect of possible disturbances leading to changes in total concentration and HMW content in the feed has not been addressed in this study. However, such variations are expected to be reduced in the future by the implementation of better process control technologies during fermentation, capture, and virus inactivation. Furthermore, these give real-time insight into product quantity and quality before or during the polishing step, thus allowing the adjustment of the load amount to avoid pool contamination. Alternatively, the off-line determination of total concentration and HMW content after virus inactivation is an option to adjust parameters before further processing.

ACKNOWLEDGMENTS

This work has been supported by the KTI (CTI)-Program of the Swiss Economic Ministry (Project 19190.2 PFIW-IW). The authors would like to acknowledge Dr. Michael Schulte and his team for advice and material during the work, and Dr. Moritz Wolf for the production of cell culture supernatant.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

ORCID

Sebastian Vogg http://orcid.org/0000-0002-6890-6744
Massimo Morbidelli http://orcid.org/0000-0002-0112-414X

REFERENCES

Arosio, P., Barolo, G., Müller-Späth, T., Wu, H., & Morbidelli, M. (2011). Aggregation stability of a monoclonal antibody during downstream processing. *Pharmaceutical Research*, *28*, 1884–1894. https://doi.org/10.1007/s11095-011-0416-7

Arosio, P., Rima, S., & Morbidelli, M. (2013). Aggregation mechanism of an IgG2 and two IgG1 monoclonal antibodies at low pH: From oligomers to larger aggregates. *Pharmaceutical Research*, *30*, 641–654. https://doi.org/10.1007/s11095-012-0885-3

Baur, D., Angelo, J. M., Chollangi, S., Xu, X., Müller-Späth, T., Zhang, N., … Morbidelli, M. (2018). Model assisted comparison of protein A resins and multi-column chromatography for capture processes. *Journal of Biotechnology*, *285*, 64–73. https://doi.org/10.1016/j.jbiotec.2018.08.014

Brown, A., Bill, J., Tully, T., Radhamohan, A., & Dowd, C. (2010). Overloading ion-exchange membranes as a purification step for monoclonal antibodies. *Biotechnology and Applied Biochemistry*, *56*, 59–70. https://doi.org/10.1042/BA20090369

Carta, G., & Jungbauer, A. (2010). Downstream processing of biotechnology products, *Protein Chromatography* (pp. 1–55). Weinheim, Germany: Wiley-VCH Verlag GmbH & Co. KGaA. https://doi.org/10.1002/9783527630158.ch1

Fargues, C., Bailly, M., & Grevillot, G. (1998). Adsorption of BSA and hemoglobin on hydroxyapatite support: Equilibria and multicomponent dynamic adsorption. *Adsorption*, *4*, 5–16. https://doi.org/10.1023/A:1008822918494

Garke, G., Hartmann, R., Papamichael, N., Deckwer, W.-D., & Anspach, F. B. (1999). The influence of protein size on adsorption kinetics and equilibria in ion-exchange chromatography. *Separation Science and Technology*, *34*, 2521–2538. https://doi.org/10.1081/SS-100100788

Hargreaves, B. (2018). Montpelier, France: William Reed. https://www.biopharma-reporter.com/Article/2018/09/11/MilliporeSigma-claims-first-of-its-kind-resin-improves-efficiency-across-the-board

Hari, S. B., Lau, H., Razinkov, V. I., Chen, S., & Latypov, R. F. (2010). Acid-induced aggregation of human monoclonal IgG1 and IgG2: Molecular mechanism and the effect of solution composition. *Biochemistry*, *49*, 9328–9338. https://doi.org/10.1021/bi100841u

Hunter, A. K., & Carta, G. (2001). Effects of bovine serum albumin heterogeneity on frontal analysis with anion-exchange media. *Journal of Chromatography A*, *937*, 13–19. https://doi.org/10.1016/S0021-9673(01)01301-2

Ichihara, T., Ito, T., & Gillespie, C. (2019). Polishing approach with fully connected flow-through purification for therapeutic monoclonal antibody. *Engineering in Life Sciences*, *19*, 31–36. https://doi.org/10.1002/elsc.201800123

Ichihara, T., Ito, T., Kurisu, Y., Galipeau, K., & Gillespie, C. (2018). Integrated flow-through purification for therapeutic monoclonal antibodies processing. *mAbs*, *10*, 325–334. https://doi.org/10.1080/19420862.2017.1417717

Jagschies, G., Lindskog, E., Lacki, K. & Galliher, P. (Eds.), 2018). *Biopharmaceutical Processing* (1st ed., pp. 3–31). Amsterdam: Elsevier. https://doi.org/10.1016/C2014-0-01092-1

Karst, D., Steinebach, F., & Morbidelli, M. (2018). Continuous integrated manufacturing of therapeutic proteins. *Current Opinion in Biotechnology*, *53*, 76–84. https://doi.org/10.1016/j.copbio.2017.12.015

Knudsen, H. L., Fahrner, R. L., Xu, Y., Norling, L. A., & Blank, G. S. (2001). Membrane ion-exchange chromatography for process-scale antibody purification. *Journal of Chromatography A*, *907*, 145–154. https://doi.org/10.1016/S0021-9673(00)01041-4

Liu, H. F., Ma, J., Winter, C., & Bayer, R. (2010). Recovery and purification process development for monoclonal antibody production. *mAbs*, *2*, 480–499. https://doi.org/10.4161/mabs.2.5.12645

Liu, H. F., McCooey, B., Duarte, T., Myers, D. E., Hudson, T., Amanullah, A., … Kelley, B. D. (2011). Exploration of overloaded cation exchange chromatography for monoclonal antibody purification. *Journal of Chromatography A*, *1218*, 6943–6952. https://doi.org/10.1016/j.chroma.2011.08.008

Reck, J., Pabst, T., Hunter, A., & Carta, G. (2017). Separation of antibody monomer-dimer mixtures by frontal analysis. *Journal of Chromatography A*, *1500*, 96–104. https://doi.org/10.1016/j.chroma.2017.04.014

Stone, M. T., Cotoni, K. A., & Stoner, J. L. (2019). Cation exchange frontal chromatography for the removal of monoclonal antibody aggregates. *Journal of Chromatography A*, *1599*, 152–160. https://doi.org/10.1016/j.chroma.2019.04.020

Tao, Y., Chen, N., Carta, G., Ferreira, G., & Robbins, D. (2012). Modeling multicomponent adsorption of monoclonal antibody charge variants in cation exchange columns. *AIChE Journal*, *58*, 2503–2511. https://doi.org/10.1002/aic.13718

Vlasak, J., & Ionescu, R. (2011). Fragmentation of monoclonal antibodies. *mAbs*, *3*, 253–263. https://doi.org/10.4161/mabs.3.3.15608

Vogg, S., Wolf, M. K. F., & Morbidelli, M. (2018). Continuous and integrated expression and purification of recombinant antibodies. *Methods in Molecular Biology*, *1850*, 147–178. https://doi.org/10.1007/978-1-4939-8730-6_11

Weinbrenner, W. F., & Etzel, M. R. (1994). Competitive adsorption of α-lactalbumin and bovine serum albumin to a sulfopropyl ion-exchange membrane. *Journal of Chromatography A*, *662*, 414–419. https://doi.org/10.1016/0021-9673(94)80530-X

Wolf, M. K. F., Closet, A., Bzowska, M., Bielser, J.-M., Souquet, J., Broly, H., & Morbidelli, M. (2019). Improved performance in mammalian cell perfusion cultures by growth inhibition. *Biotechnology Journal*, *14*, 1700722. https://doi.org/10.1002/biot.201700722

Wollacott, R., Roth, L., Sears, T., Sharpe, R., Jiang, M., & Ozturk, S. (2015). The development of a flow-through mode cation exchange process for the purification of a monoclonal antibody. *Bioprocessing Journal*, *14*, 5–13. https://doi.org/10.12665/J142.Wollacott

Wrzosek, K., & Polakovič, M. (2011). Effect of pH on protein adsorption capacity of strong cation exchangers with grafted layer. *Journal of Chromatography A*, *1218*, 6987–6994. https://doi.org/10.1016/j.chroma.2011.07.097

Zhou, J. X., & Tressel, T. (2006). Basic concepts in Q membrane chromatography for large-scale antibody production. *Biotechnology Progress*, *22*, 341–349. https://doi.org/10.1021/bp050425v

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Vogg S, Pfeifer F, Ulmer N, Morbidelli M. Process intensification by frontal chromatography: Performance comparison of resin and membrane adsorber for monovalent antibody aggregate removal. *Biotechnology and Bioengineering*. 2020;117: 662–672. https://doi.org/10.1002/bit.27235